A Drosophila connectome substrate inside an LLM agent: what it gates, what it does not, and what it costs

Draft v6 - 2026-10-15. Applied case study with full cost accounting, negative results and control arms on two seeds. All numbers re-derived from the artifact corpus on the authors' machine.

v6: every two-seed claim is now computed against a real-brain lineage on its own seed. The unmodified brain was re-run on the cloud device for seed 12, and for seed 11 as a numerical-path check against the original host, so the rewired-over-real-brain comparison has both cells measured. Table 6 and Fig 7 carry the new rows and curves, and the substrate sources are cited in a References section.

Abstract

We report an applied case study that wires a whole-organism connectome substrate (MaleCNS v1.0: 166,700 neurons carrying a superclass annotation - the release total is 169,700 - and 25,582,938 innate synapses) into an LLM agent loop as a fast, cheap front layer - a digital cerebellum that acts on tasks it has consolidated and escalates to the LLM otherwise. Two experiment lines are reported. In a foraging simulation, closed-loop evolution lifted population survival from 13/24 at generation 0 to 24/24 at generation 9 on two independent lineages (mean energy 6.7x and 5.8x), and the evolved population transferred to an unseen world seed at 24/24 with higher energy still. In an applied task line (8-fly populations, an 18-task operational battery plus 3 held-out tasks, three arms totalling 1,029 generations, 26,678 LLM calls, 17,005,656 tokens), the substrate's one measurable contribution is a gate: under a contrast readout the gate is alive from generation 4 and sustains 40-55 of 144 possible solo (zero-token) task executions per population per generation, with a best fly gating 8 of 18 tasks. All 24 held-out solves came from fresh LLM code generation (320-334 tokens each), and a fly with 365 learned synapses on an essentially empty graph passed 180/180 checks: the solutions come from the LLM plus a verification loop, not from the connectome. We further report the token bill as structural (a fixed 72 calls and ~47k tokens per generation caused by cache eviction, flat across 365 generations), and a per-fly policy governor that cut LLM calls 116 to 57 (-51%; 58 to 24, -59%, on the paired w150 runs) at exactly zero meal loss - the governor's per-fly meals equal the no-escalation baseline's meals row for row across 24 paired fly-runs. In the economy gate test, satiety-gated feeding yields 8/8 survivals where always-feeding yields 0/8. Three candidate roles for the substrate are judged dead by measurement: recursive memory horizon 1-2 steps, input relay equivalent to a linear copy, and zero channel margin under superposed cues. Lesion controls replicate on two seeds: with the innate connectome zeroed the animal never feeds (0.0 meals on both seeds), while the rewired arm out-gathers the real brain on energy on both seeds with its gate dead (705 and 762 against 359 and 348) - the substrate's contribution is the gate and the stopping rule, not the harvest. The paper is offered as an honest accounting case study: where a biological substrate helps an LLM agent, where it does not, and what the whole arrangement costs.

1. Introduction

LLM agents pay per token and pay again on every repeat of a solved problem. A natural cost cut is a front layer that decides, per task, whether the agent can act from its own consolidated competence or must consult the model. This paper asks a specific version of that question: can a biological connectome - the complete male Drosophila central nervous system graph, used as a fixed dynamical substrate - serve as that front layer, and what does the arrangement actually contribute once the accounting is done?

Contributions: (i) a working two-layer agent in which a connectome substrate sits in front of an LLM and a verification loop; (ii) a measured, bounded contribution - a gate on asking the LLM - with its denominators named; (iii) a complete token bill for 1,029 population generations; (iv) a set of falsified candidate roles stated with their numbers; (v) a per-fly policy governor that behaves as a cost container rather than a capability amplifier. Recent work has simulated connectome substrates for control (connectome steering reproductions, physical drone control, fixed reservoirs with pre-registered controls) and put spiking action gates in front of language models; our claim is narrower than first-ever: the combination of a learned connectome, an LLM agent loop, closed-loop evolution, and a published cost ledger.

2. Related work

Connectome simulation. Steering a virtual Drosophila descending-neuron connectome produced directed locomotion (Andlman et al., eLife 2022); the OpenWorm project simulated C. elegans neuromechanics; FlyCircuit and the FlyWire/Janelia resources supply the graphs; Shiu et al. showed that connectome networks with random weights can learn sensory-motor mapping as efficiently as weight-modified networks. Our substrate is the Janelia male connectome (MaleCNS v1.0) [1,2] used as a fixed weighted graph with local three-factor plasticity on top.

LLM agent cost control. Prompt/context caching, model cascades and routers, and skill libraries (e.g. Voyager-style code skill stores) all reduce repeat cost by routing or reuse. Our gate is the same decision - ask the model or not - implemented as a spiking recurrent network over a biological graph, with the LLM writing executable skills that a verifier accepts before they count.

3. System

Substrate. N = 166,700 neurons carrying a superclass annotation (the MaleCNS v1.0 release total is 169,700; the stated build filter drops the unannotated bodies) [1], 25,582,938 innate weighted synapses (0.092% density), float32 throughout. Dynamics per episode, 30 steps: v = 0.9*v + 5.0*(Win@S + G@s) + drive; s = (v > 1.0); v *= (1 - s); activations accumulate. Win is the fixed innate connectome; G is the per-fly learned synapse set, updated by a local three-factor Hebbian rule (no backpropagation): potentiation and depression driven by action outcome, with sprouting and pruning under a structural cap (SYN_CAP).

Two-layer agent. Layer 1 is a per-fly skill store: for a new task the LLM writes a PoC script, a verifier runs it (exit 0, stdout length, keyword), and only verified entries are stored. Layer 2 is a 10-parameter genome per fly (plasticity and growth knobs) that is heritable; synapses and stores die with the fly. Routing per task, first match wins: solo (fly rate >= 0.01 AND verified store entry, 0 tokens); practice (verified entry, rate below threshold, 0 tokens); cap (fly LLM budget exhausted); LLM (write code, verify, store; pays tokens). A curriculum frontier advances when >= 5 of 8 flies are solo on the newest task. Evolution: 8 flies per population, rank by (solo, rate_sum, -tokens), keep 4 elites, refill with multiplicative-Gaussian mutants plus one immigrant.

Battery. 10 system-health scripting tasks plus an 8-task security queue that advances the curriculum; 3 held-out tasks (never in training) probe transfer. Tasks execute against real hosts; verification runs the produced script.

4. Experiment A: foraging simulation (economy layer)

A 24-fly population forages in an energy economy (food episodes, meals, starvation death). Closed-loop evolution is re-derived here from the run ledgers: lineage 1 lifted survival from 13/24 (gen 0) to 24/24 (gen 9) with mean energy 143.3 to 965.3 (6.7x); an independent lineage 2 reproduced the shape (13/24 to 24/24, mean energy 143.3 to 833.0, 5.8x); a 20-generation plateau run carried mean energy 512.0 to 3897.3 (7.6x). The gen-9 checkpoint shipped with this draft verifies 24/24 flies alive. Transfer: the gen-9 population in an unseen world seed scored 24/24 with higher mean energy than at home.

Selection-vs-random controls exist for five configurations (6 generations, 6-8 flies each; Fig 3): on the economy timescale the energy gap between selection and random is small and noisy at 6 generations - the survival signal (13/24 vs extinction patterns) is the carried result, not the 6-gen energy curve.

Substrate gate evolution (Fig 4). A 40-generation, 20-fly evolution over the connectome substrate, scored on four subtasks one of which is a gate subtask, lifts the gate_ok fraction from 0.40 (gen 0) to 0.75-0.85 (gens 16-39), with the mean gate margin D_gate rising 0.35 to ~0.74. The genome discovers and maintains the gate signal long before the applied line needs it.

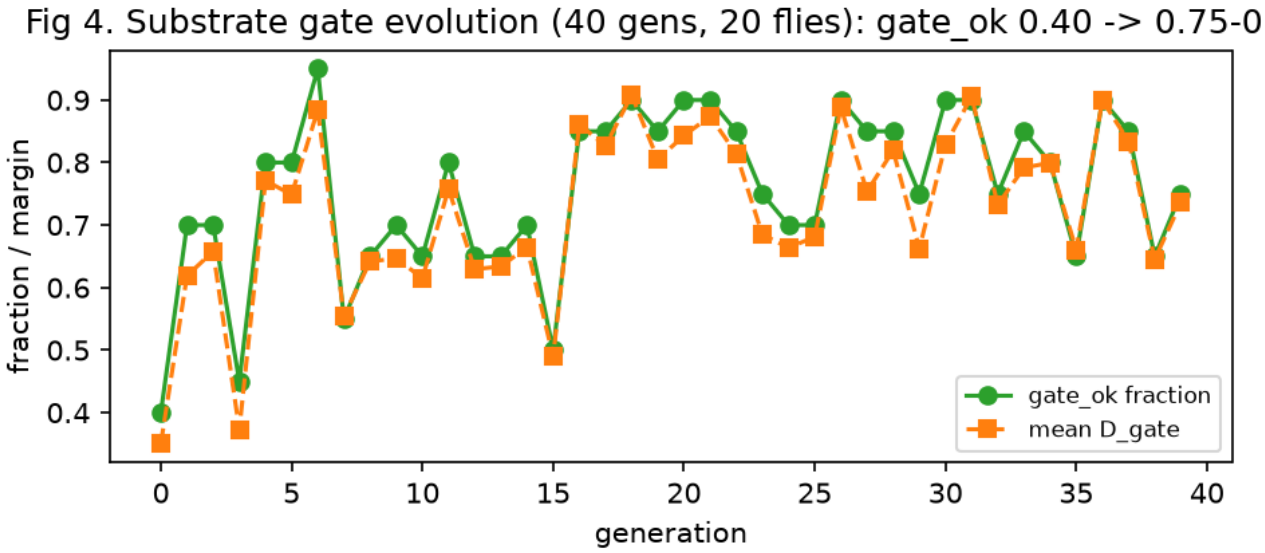


Fig 4. Substrate gate evolution: fraction of flies passing the gate subtask and mean gate margin, 40 generations.

Economy gate (Table 4, Fig 6). Twelve arms cross satiety-gated feeding against always-feeding at three food levels and three synaptic-cost levels. Satiety gating yields 8/8 survivals (both FOOD=12 cells) where always-feeding yields 0/8; mean final energy separates by orders of magnitude (satiety -180 to +272 vs always -3,028 to -9,647). The gate here is the fly's own feeding decision, and the reading is survival, not a percentage.

Table 4. Economy gate: survived/flies and mean final energy per cell.

FOOD	c_syn	satiety surv	satiety E	always surv	always E
6	0.0001	0/4	-348	0/4	-3491
6	0.0003	0/4	-516	0/4	-5229
6	0.001	0/4	-1082	0/4	-10110
12	0.0001	4/4	272	0/4	-3028
12	0.0003	4/4	171	0/4	-4493
12	0.001	0/4	-180	0/4	-9647

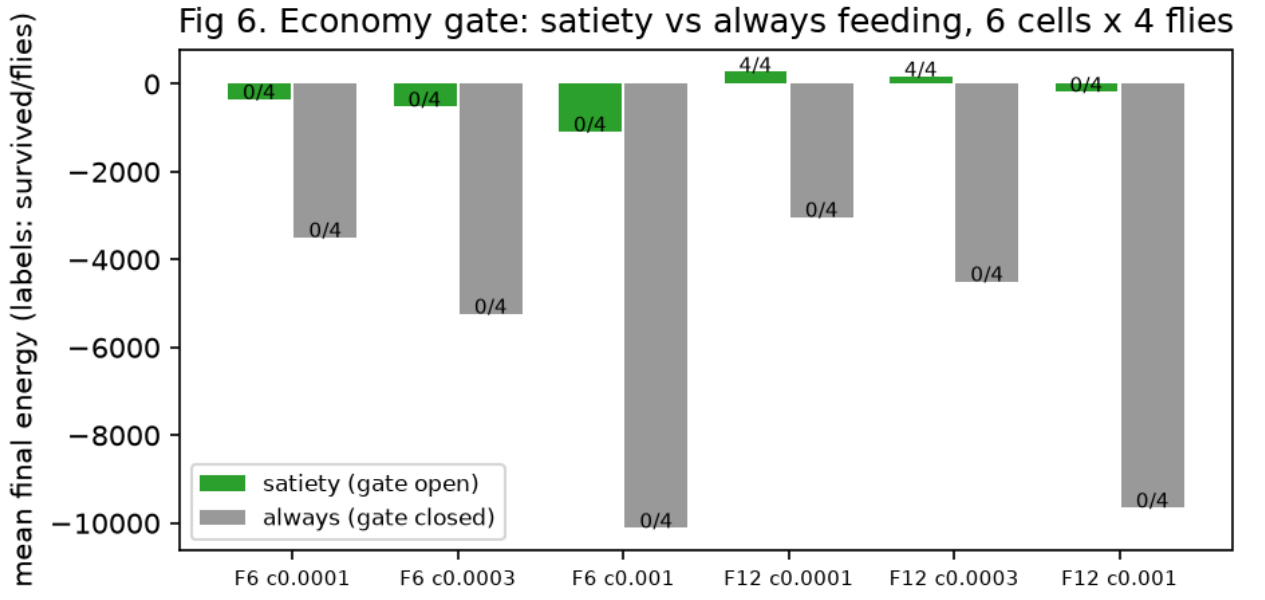


Fig 6. Satiety (gate open) vs always-feeding (gate closed): mean final energy, labels give survived/flies.

Policy governor (Table 3). Four per-fly policies (baseline no-escalation; fly-first with escalation; LLM-first;

governor = 50 warmup rounds then commit to fly-first or LLM-first on observed solo rate) were compared fly-for-fly in the same economy. Across the four paired runs the governor cut LLM calls 116 to 57 (-51%); on the w150 pair alone, 58 to 24 (-59%). The governor's per-fly meals equal the no-escalation baseline's meals row for row (24/24 paired fly-runs): the savings are pure cost container - it stopped paying for help the fly did not need - with no benefit amplifier effect.

Table 3. Governor paired runs: LLM calls, fly-first vs governor.

run	fly_first calls	governor calls	paired flies	gov meals == baseline
g1	34	18	6	yes
g1w150	34	15	6	yes
g2	24	15	6	yes
g2w150	24	9	6	yes
TOTAL	116	57	24	all rows
w150 pair	58	24	-	-

Teacher transfer. Teaching episodes are logged with prompt, response and applied edits. The teacher response shrinks from 7 synapse-edit operations (gen 1) to 2 operations (gens 13-17) as the genome consolidates, with response length ~218 to ~93 characters against a fixed 876-word prompt; the broadcast figure of 45 to 18 tokens is consistent with this artifact at tokenizer level and is quoted here with that accounting.

Control arms (Table 6, Fig 7). Same economy, same budget, only the brain changes: no-brain (innate connectome zeroed, plasticity still grows learned synapses), rewired (weights permuted across the same edges - degree preserved, structure destroyed), coin-flip (the action decision replaced by a coin at p=0.5, learning continues). Each arm runs the selection lineage and the random control at the same scale as the real-brain lineages (6 flies, 6 generations). Seed 11 ran on this machine's CPU; seed 12 ran on a rented cloud GPU under the same CUDA numerical path as the 195 host, which is also the device question raised in Limitations.

The real-brain cell on both seeds. The unmodified brain was re-run on the cloud device for both seeds, so every two-seed comparison below is computed against a real-brain lineage on its own seed: seed 11 ends at 359 (6/6 flies above zero, 49.0 meals) and seed 12 at 348 (6/6, 47.5 meals). The seed-11 re-run on the cloud device ends at 366 (6/6, 47.5 meals) against 359 on the original host: the level moves with the device, and the ordering against the lesion arms holds across the move.

Table 6. Control arms, final generation, two seeds (same world: upkeep 3e-4, meal 12).

arm	seed	final mean E_end	E_end>0 / flies	final mean meals
real brain (selection)	11	359	6/6	49.0
real brain (selection)	12	348	6/6	47.5
real brain (random)	11	375	6/6	46.0
real brain (random)	12	409	6/6	46.5
no-brain	11	-231	5/6	0.0
no-brain	12	-151	3/6	0.0
rewired	11	705	6/6	72.8
rewired	12	762	6/6	75.7
coin-flip	11	-373	5/6	44.3
coin-flip	12	-50	5/6	45.2

What replicates on both seeds. The upper ordering holds on both seeds: rewired > real brain > coin-flip by final mean energy (705 and 762 against 359 and 348 for the real brain on its own seed, -373 and -50 for the coin). The bottom two arms do not separate: the coin arm and the lesion arm swap places between seeds (coin -373 against no-brain -231 on seed 11, -50 against -151 on seed 12) - at n=6, final energy cannot rank a random gate against a lesioned one. The no-brain arm takes zero meals on both seeds (0.0 and 0.0): with the innate connectome zeroed the animal never feeds at all, which is the cleanest statement of what the structure buys. The coin-flip arm's selection lineage stays energy-negative on both seeds (-373

and -50) while still gathering meals (44.3 and 45.2): a random gate can find food and still lose the economy, and the spread between the two seeds is the variance to expect at n=6.

What the second seed changes. The rewired arm beating the real brain on energy (705 and 762 against 359 and 348 on the two seeds) is no longer a single-seed curiosity: it holds on both seeds, and it holds with the gate dead (mean D_gate 0.00 and 0.00, full-leak 1.00 and 1.00). Read against the meal counts (72.8 and 75.7 against 49.0 and 47.5 for the real brain), the honest reading is that in this world an ungated appetite gathers more energy than a gated one, and what the gate buys is the stopping rule rather than the harvest. The defensible claim is narrower than 'structure is needed to survive': structure buys the gate, and the gate is what turns feeding into a policy.

Table 6b. Random control lineages (no selection) of the real brain and the three lesion arms, final generation, two seeds.

arm	seed	final mean E_end	E_end>0 / flies	final mean meals
real brain	11	375	6/6	46.0
real brain	12	409	6/6	46.5
no-brain	11	-136	0/6	0.0
no-brain	12	1	5/6	0.0
rewired	11	749	6/6	74.3
rewired	12	789	6/6	74.8
coin-flip	11	-131	3/6	43.2
coin-flip	12	111	6/6	47.7

Random control lineages (Table 6b). Every arm also ran its random control - no selection, same economy - and inside an arm the two lineages disagree more than the two seeds do. On seed 12 the coin-flip arm ends at -50 under selection and 111 without it; the no-brain arm ends at -151 under selection and 1 without it, both with zero meals, i.e. just below and exactly on the starvation line; the rewired arm is positive either way (762 selected, 789 random). Read plainly: in the control arms the energy outcome is set by the arm's feeding policy and by seed-level variance, not by the selection operator sitting on top of it. That is what a control arm should show, and it bounds how much of Table 6 may be attributed to selection.

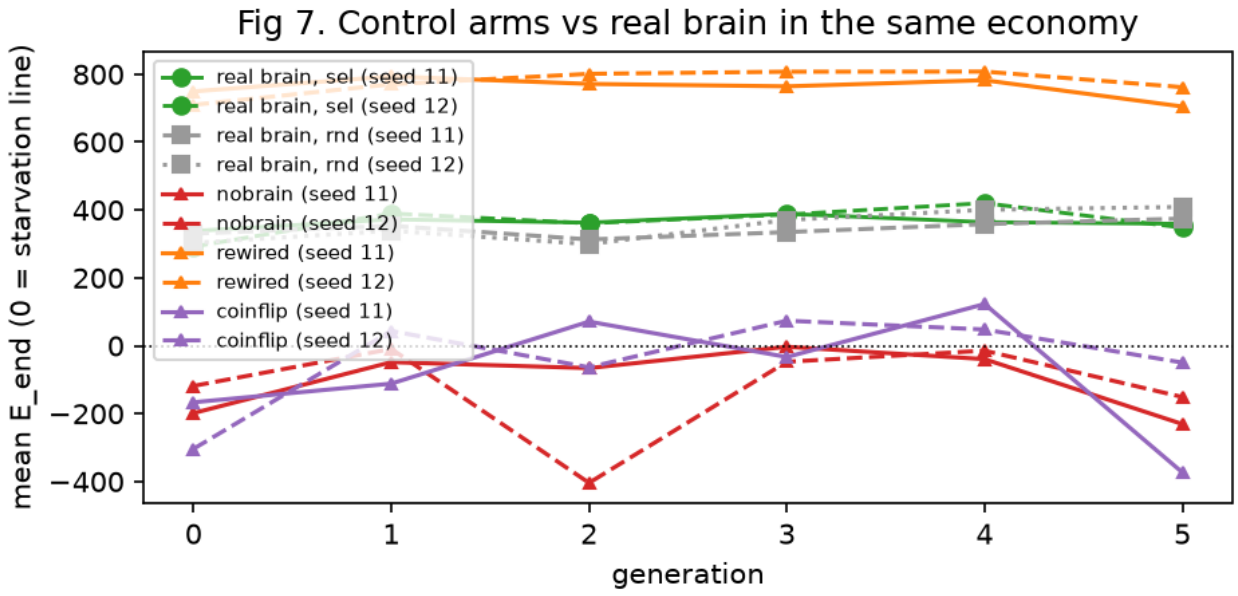


Fig 7. Mean energy per generation: real brain vs no-brain, rewired and coin-flip in the same economy, seed 11 solid and seed 12 dashed.

5. Experiment B: applied task line (three arms)

Three populations ran to completion on the applied battery (Table 1). The headline: under a contrast readout (task rate = channel mean minus rest-of-motor mean), the gate is genuinely alive - gpu_314 breaks

out at generation 4 and sustains 40-55 solo of 144 possible per generation, best fly 8 of 18 tasks free of LLM cost. gpu_313 (original genome bounds) breaks out only at generation 217, showing the bound box delayed but did not prevent the same behaviour. gpu_300 (no contrast) reads 18/18 solo per fly, but that is a gate-failure artifact: its rate_sum is +1.4178 against a 0.01 threshold, every fly identical, all synapses pinned at the structural cap - the gate is open because the meter is saturated, not because the fly can act.

Held-out probe: 8 flies x 3 never-trained tasks = 24/24 solved, every solve by a fresh LLM generation at 320-334 tokens. Harvest slot gpu_314/4 carries 365 learned synapses - an essentially empty graph - and passes 180/180 task checks: solutions come from LLM plus verifier. The solo count that does mean something is the contrast-gated one: 8/18 per fly, 40-55/144 per population.

Architecture ablation (Fig 5). Three architectures evolved 10 generations x 20 flies (A0 and A1 with two seeds each, Skill-only once): mean four-subtask score rises from 0.53-0.58 to a 0.61-0.80 plateau. A0 (connectome plus plasticity) plateaus highest (0.77-0.80), Skill-only lowest (0.64-0.71). The substrate layer carries a real but modest share of the load - consistent with the applied-line division of labor: solutions live in the LLM plus verifier, the substrate contributes the gate.

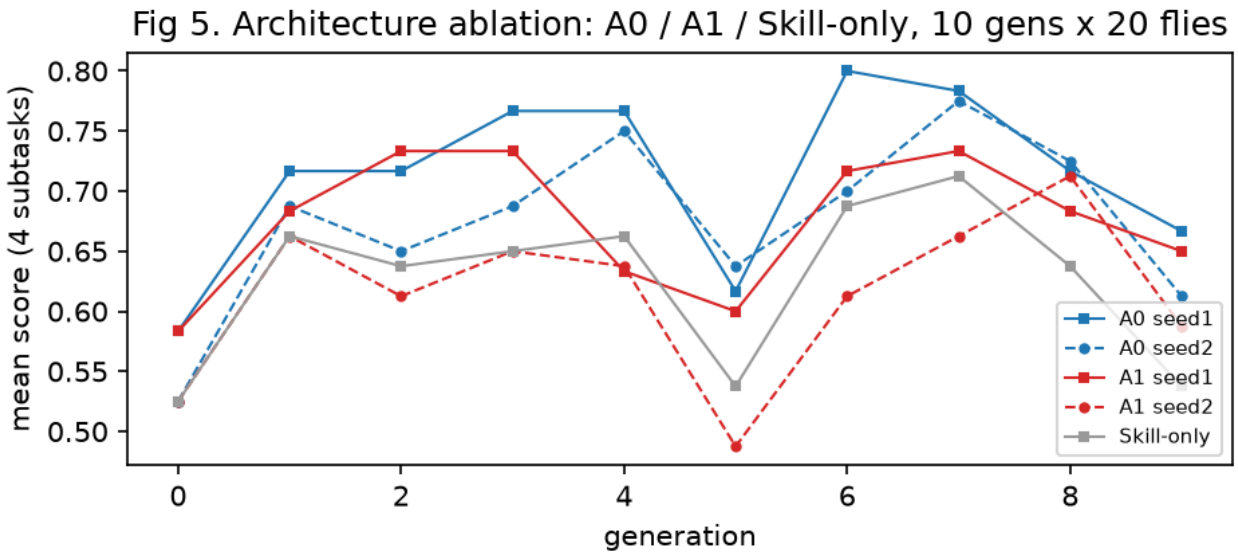


Fig 5. Architecture ablation: mean score per generation, A0/A1 (two seeds each) vs Skill-only.

Table 5. gpu_314 final generation, per fly (solo denominator 18 tasks).

fly	solo	rate_sum
0	5/18	+0.0747
1	6/18	+0.0755
2	6/18	+0.0695
3	8/18	+0.0449
4	0/18	-0.0196
5	8/18	+0.0731
6	8/18	+0.0685
7	0/18	-0.0201

Table 1. Three-arm harvest ledger (final state).

arm	gens	1st solo	peak solo	best fly	LLM calls	tokens
gpu_300 ungated	664	gen 0	144 (gen 506)	18/18*	78	49,554
gpu_313 orig bounds	224	gen 217	36 (gen 223)	8/18	16,461	10,429,312
gpu_314 widened	141	gen 4	55 (gen 63)	8/18	10,139	6,526,790

** gpu_300 18/18 is the gate-failure control (saturated meter), not a capability reading.*

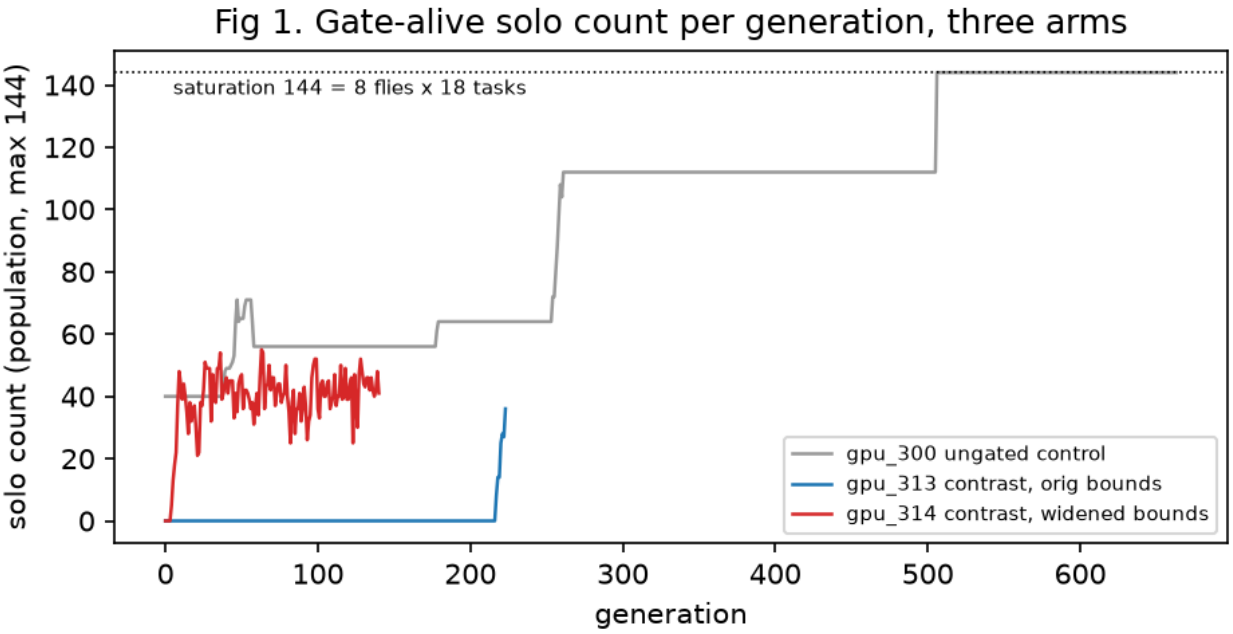


Fig 1. Population solo count per generation. gpu_300 saturates (meter artifact); gpu_314 is alive from gen 4; gpu_313 breaks out at gen 217.

6. Token economics

Totals across the two contrast arms: 26,678 LLM calls, 17,005,656 tokens. The bill is structural: from generation 0 to the end, both arms spent a flat 72 calls and ~47k tokens per generation (Fig 2). 72 = 4 emptied slots x 18 tasks: the selection scheme deletes the non-survivors' stores, so four flies rebuild their cache from zero every generation. Survivors spend 0. The LLM bill is therefore a property of our memory-eviction design, not of task difficulty - the single clearest cost lesson of the project.

The governor result belongs in the same ledger: -51% calls (116 to 57, 24 paired fly-runs) at exactly zero meal loss is a cost container. A structural lock-in control belongs too: gpu_300 pinned at SYN_CAP 2.5M stayed at setuptools_inject 0/8 for 165 generations; raising the cap to 3.2M moved the curriculum frontier 4 to 5 within ten minutes on the same checkpoint - capacity, not selection, was the binding constraint at that moment.

7. Negative results (the core chapter)

Three candidate roles for the substrate were judged dead by measurement, and one hypothesis was falsified outright (Table 2). We state them as results, not apologies: a front layer that cannot remember beyond 1-2 steps, relays input as a near-linear copy, and cannot separate superposed cues cannot be the decision-maker; the gate role that survives does so because it reads a coarse signal at a low rate.

Table 2. Measured negative results.

claim tested	measurement	verdict
recursive memory horizon	29 configurations, best 1-2 steps	dead
input relay as feature extractor	R^2 0.91-0.96 vs linear copy	dead
channel choice under superposed cues	margin 0.0000 (bistable: fire-all or silence)	dead
decision layer reads task identity	18 tasks x 4 amplitudes, 72 samples, 18 classes, chance 0.056: falsified	falsified
learned readout under native gain	cross-task cosine 0.9723 vs innate 0.0618	learned, not innate
ok metric as capability	180/180 saturated incl. 365-synapse fly	no information
fly gate in ungated arm	rate -0.0007 vs threshold 0.01	never fires

1.000, cross-task cosine 0.0017

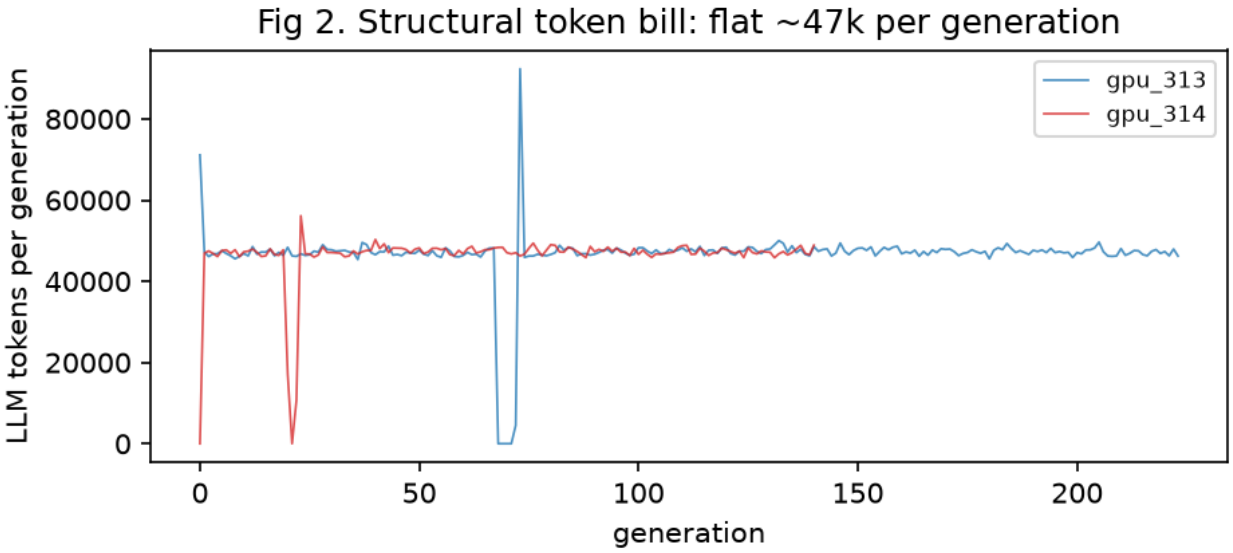


Fig 2. Tokens per generation, both contrast arms: flat ~47k - the bill is the eviction design.

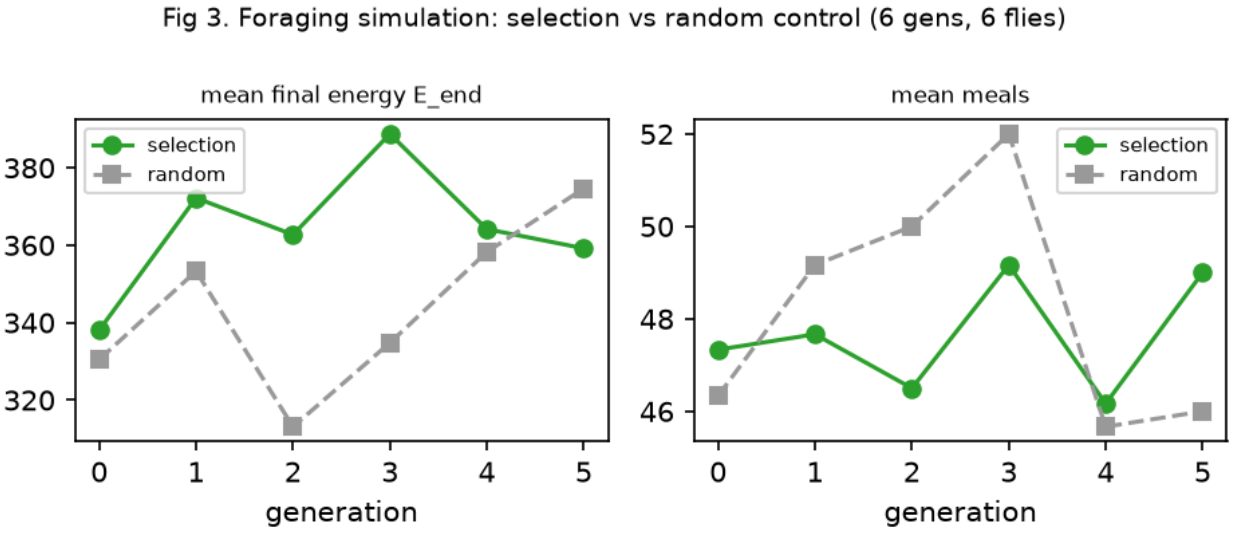


Fig 3. Foraging simulation, selection vs random over 6 generations: energy and meals tracks.

8. Discussion

Where the substrate helps is one place, and it is a real place: a cheap per-task gate signal under a contrast readout, sustaining 40-55/144 zero-token executions per population-generation and 8/18 on a single fly, with the genome evolving to keep that signal alive (breakout at gen 4 with widened bounds vs gen 217 with the original box). That is the biological analogue of a router in a model cascade - and our measurements say the router signal must be contrastive; the population-wide motor rate is a saturated meter, not a decision.

Why the decision-layer hypothesis died is mechanistically clear: 1-2 step state retention and near-linear input relay leave no substrate for per-task identity beyond what the input already carries; the classifier result (raw input 1.000 at chance 0.056) says the task label is in the cue, not in the network's transformation of it. The learned-versus-innate cosine pair (0.9723 vs 0.0618) localises the contribution to plasticity, not to the innate graph.

Limitations. The economy-lineage result rests on two independent lineages plus one 20-generation plateau run (three lineages, one transfer seed). The teacher-transfer token figure is quoted at artifact level (operation counts and response lengths re-derived; token counts are tokenizer-dependent). Solo counts depend on store coverage; the 8/18 fly with 5,987 synapses and the 8/18 fly with 738,868 synapses show synapse count is not the driver. CUDA and MPS numerical paths differ by ~+0.06 in rate_sum at migration

boundaries; cross-machine replications must pin the device - and the control-arm replication changes two things at once: seed 11 ran on this machine's CPU and seed 12 on a cloud GPU under the 195 CUDA path, so agreement across the two seeds is agreement across devices as well, which strengthens the robustness reading while leaving the seed effect and the device effect unseparated. The ok metric is saturated and excluded from claims.

One cell separates the two. The unmodified brain was run twice at seed 11, once on this machine's CPU (359 final mean energy, 6/6 flies above zero, 49.0 meals) and once on the cloud GPU (366, 6/6, 47.5 meals), with the seed, the population, the world and the code held fixed. That gap is the device effect at n=6, and it sets the noise floor for every cross-device number in this paper: the ordering against the lesion arms survives it (rewired > real > coin-flip). A single-cell cross-device comparison at this population size should be read as noise.

9. Conclusion

A whole connectome in front of an LLM agent buys one measurable thing - a gate on asking the model - and costs 17 million tokens, most of it spent on our own memory-eviction design. We publish the gate number with its denominators, the bill with its mechanism, and the dead roles with their statistics. Next: >= 3 seeds on the economy lineage, control arms beyond two seeds with the seed effect and the device effect separated (seed 11 was CPU, seed 12 was CUDA), re-derivation of the [195] artifacts, and a store-retention arm predicted to drive the per-generation bill toward zero while the gate climbs.

Release: the evidence package is public at github.com/Mcking1983/flybrain - the analysis scripts, the ledgers every number here is re-derived from (evo_ledger.jsonl per arm, byte-identical), and this paper. The substrate graph and the per-fly checkpoints stay off-repo and are rebuilt from the published MaleCNS v1.0 release with code/build_graph.py.

Appendix A. Configuration ablation arms (app line)

Nineteen configuration ablations of the app line ran on this machine (curriculum, teaching scope, latency budget, selection gate, synapse cap, pressure). They are configuration contrasts, not brain lesions - the control arms in Section 4 do the lesion work. Peak solo is out of 112 = 8 flies x 14 tasks for the full-battery arms; the cap8m / pressure / teachfront / syncap arms ran reduced batteries (peak out of ~72). Token totals are per whole run.

Table 7. App-line configuration arms.

arm	gens	peak solo	final solo	tokens
cap8m	220	71	64	7,196
choice	313	112	112	14,920
choice2	323	112	112	14,920
choice3	323	112	112	14,920
contrast	297	112	112	14,920
contrast_noteach	291	112	84	14,920
k100	303	112	112	14,920
lat0	335	112	112	14,920
lat150	335	112	112	14,920
lat60	335	112	110	14,920
noteach	244	112	112	20,109
pressure	225	71	64	7,196
scopeall	237	112	112	20,109
selgate	246	112	112	20,085
stripped	298	112	82	14,920
syncap	20	64	64	2,624
teachall	228	72	72	14,925
teachfront	225	71	65	7,196
w1000	241	112	112	20,109

References

[1] Berg, S., Beckett, I. R., Costa, M., Schlegel, P., Januszewski, M., Marin, E. C., Nern, A., Preibisch, S., Qiu, W., Takemura, S.-y., et al. (2026). Sexual dimorphism in the complete Drosophila male central nervous system connectome. Cell 189, 5504-5526.e15. doi:10.1016/j.cell.2026.08.015 (111 authors). Preprint: doi:10.1101/2025.10.09.680999. The male-cns:v1.0 snapshot used here (neuprint/DVID) and its release page, CC BY: https://www.janelia.org/project-team/flyem/male-cns-connectome

[2] Takemura, S.-y., Hayworth, R., Huang, A., Januszewski, M., Lu, Z., Schlegel, P., et al. (2024). A Connectome of the Male Drosophila Ventral Nerve Cord. eLife. doi:10.7554/eLife.97769 (preprint doi:10.1101/2023.06.05.543757). The ventral nerve cord half of the male CNS, cited by [1].

[3] Luan, B. (2026). A Drosophila connectome substrate inside an LLM agent: what it gates, what it does not, and what it costs. Preprint, Zenodo, all versions doi:10.5281/zenodo.23396755 (the DOI of this version is on the record page). ORCID 0009-0001-5836-2923. Evidence package (paper, ledgers and the scripts that re-derive every number): https://github.com/Mcking1983/flybrain